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Kumsong Jo, Sunhak Kim, and Bongnam Ri

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Abstract

This paper proposes an Exponentially Weighted Moving Mean–Variance (EMMV) model that discounts older data with a forgetting factor. Unlike traditional asset-level smoothing, the EMMV aggregates portfolio-level moments across rolling windows, yielding stable, adaptive allocations. Hyperparameters are selected objectively via a genetic algorithm with walk-forward cross-validation. Using 30 U.S. stocks from 2010 to 2025, the model achieves a buy-and-hold Sharpe ratio of 1.49 — more than double that of the equally weighted portfolio — and a rebalanced Sharpe ratio of 0.69 after transaction costs, outperforming all benchmarks. Robustness analysis confirms consistently high performance over a wide parameter range, and bootstrap tests verify significant or near-significant improvements. The framework provides a practical, data-driven tool for dynamic portfolio selection in nonstationary markets.

Keywords: Mean–variance (MV) model ▪ the exponentially weighted moving mean–variance (EMMV) ▪ Sharpe ratio ▪ genetic algorithm (GA)

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